

Assessment 3 - Storytelling with Open Data

Assessment 3

Storytelling with Open Data



Course Code: MATH2237/MATH2270

Weighting: This assessment is worth 40% of your final mark for this course

Length: See instructions below

Task approach: Individual

Due date and time: During Assessment Period Week 1, Wednesday 11th June 2025 23:59 AEST

Feedback: Feedback will be provided via Canvas

Please click or tap on the relevant sections to find out more:

▼ 1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

1. identify your target audience and determine a data visualisation design goal;
2. use data visualisation and storytelling techniques, including verbal, written and interactive features to help engage your audience and leave a lasting impression;
3. reflect on the major professional, ethical and integrity-based issues that arise during the practice of data visualisation;
4. apply and integrate expert-informed best practice and research knowledge to enhance the effectiveness of your data visualisations and critique the work of others;
5. conceptualise multiple data visualisation designs and determine the most appropriate strategy to achieve your goal taking into account your audience and the nature of the data;
6. source, review and prepare data required for your data visualisation; and program leading, web-based, open-source, interactive data visualisation technology to build, deploy and

disseminate your data visualisations, including applications and dashboards.

▼ 2. Overview of Assessment

In the previous assessments, you were asked to find data visualisations designed by others to create a vodcast that demonstrated your skills in storytelling techniques and publish a web report improving the work of others. In this assessment, you will work on a data visualisation project that will tell a compelling and authentic story using open data. You choose the topic, data, and the story. This is your chance to showcase your data visualisation skills.

▼ 3. Assessment Task

The goal of Assignment 3 is simple. **Tell a compelling story using data visualisation.** You can choose your own story and data or pick from one of the approved topics below. Your assignment must follow the instructions and constraints below.

- Assignment 3 must be based on open or public data. **Ensure you include a reference to your data source(s) in the web hosted assignment.**
- Your assignment must be an **R-based, web-hosted dashboard, application or interactive slideshow/storyboard**. You can use static or interactive/animated plots. Reports and video presentations are **NOT** permitted. How you design and tell the story is up to you. However, the following restrictions apply:
 - **Dashboards or applications:** One HD screen (1080p - no scrolling) with a maximum of 3 pages/areas/tabs. There is no limit to the number of visualisations you can include on each page. See [Shiny](#) ↗, [shinydashboard](#) ↗ or [flexdashboard](#) ↗ for dashboard and application development.
 - **RMarkdown slideshow/storyboard:** 10 transitions/slides max. Slides can be a combination of text (bullet points), media (use sparingly), data visualisations (which must be the focus) and a reference list. There is no limit to the number of visualisations you can include in the slideshow. I recommend [reveal.js](#) ↗ or a [flexdashboards storyboard](#) ↗ layout for HTML based RMarkdown presentation libraries.
- Your assignment must be hosted online and publicly accessible. You must use one of the following free services:
 - [RPubs](#) ↗ (Does NOT support Shiny - RMarkdown HTML outputs only)
 - [shinyapps.io](#) ↗ (Supports Shiny)
 - [Posit Connect Cloud](#) ↗ (Supports Shiny)
- You must use free services. **DO NOT pay to have your assignment hosted online.** If you have a resource-hungry app that won't run on a free service/subscription, contact the course

coordinator for assistance. This request must be made 48 hours before the assignment deadline.

Approved Topics

If you are short on time and a topic and dataset for Assignment 3 to get started, you can use one of the following approved topics.

- **Australian Federal Election 2025:** The 2025 Australian Federal Election will take place on the 3rd of May and is shaping up to be a close one. The [Australian Electoral Commission \(AEC\) website](#)  will publish the official results. This site also includes the results of all past elections and election boundary information (including spatial data). What story can you tell?
- **Crime Statistics Agency:** Tell a story about crime using [Crime Statistics Agency](#) (CSA) data. CSA is responsible for the independent analysis and reporting of crime statistics in Victoria.
- **Cost-of-living crisis/inflation:** Tell a story about the cost-of-living crisis/inflation in Australia (or around the world). There are many data sources to consider. The Consumer Price Index ([ABS](#)) report is a great place to start. You can focus your story on specific costs, e.g, housing, rent, food etc.

Please feel free to discuss and get feedback on your story ideas in [Assignment 3 - Discussion](#).

RStudio/Posit Cloud

Please see [Course resources](#) for information on accessing RStudio/Posit Cloud. It is important that you use RStudio/Posit Cloud because it will allow teaching staff to log into your projects if you need assistance.

▼ 4. Timeline

To help you prepare for the assessment, please use the following timeline as a guide:

Timeline	Activities
Week 9	• Familiarise yourself with Assignment 3 • Check the FAQs or post any questions you have on Assignment 3 - Discussion • Create a Posit Cloud account if you haven't already done so.
Weeks 10 - 11	• Finalise your topic choice (or choose one from the list of approved topics). • Source, download and start preprocessing/exploring the necessary data. • Decide how you will present Assignment 3 (Storyboard OR App OR Dashboard). • Outline a narrative using your topic and data.

Timeline	Activities
	• Continue refining your objective, audience and narrative components • Commence designing and building the data visualisations you need to deliver your narrative • If you are planning to build interactive plots or apps, ensure these features are well justified • Choose your method to publish your assignment online (e.g. Posit Cloud Connect, Shinyapps.io, RPubS etc).
Weeks 12 - 13	• Finalise your objective, audience and narrative components • Continue designing and building the data visualisations you need to deliver your narrative • Test and ensure you have time to debug interactive data visualisations and apps • Practice publishing a draft of your assignment online to ensure you understand the process (E.g. PowerPoint, Shinyapps, RPubS etc) • Ask for advice or help with R on Discussions.
Week 14	• Review and finalise your assignment. • Publish your assignment online and test accessibility and function. • Ensure you complete the Assignment 3 - Cloudflare Radar URL Scan Result Upload step. • Complete all the steps detailed in the Assessment submission instructions (see below)

▼ 5. Assessment submission and feedback

- Use Posit Cloud to create an R-based, web-hosted (e.g. Plotly, RPub, Shinyapps.io etc.) data visualisation dashboard/app/slideshow etc. (see above).
- Complete the [Assignment 3 - Cloudflare Radar URL Scan Result Upload](#) step. This is proof that the assignment has been successfully hosted online. If you do not complete this step, Assignment 3 will remain unmarked until you do it.
- You must include a reference(s) to your data source(s) in the web-hosted assignment!
- Download [Assignment 3 Template.docx](#)  and fill in the following information:
 - Student name and number
 - Assessment declaration checklist
 - URL to your assignment hosted online
 - A reference list formatted in [APA 7th Ed style](#) with your data sources and any other references used

- Copy and paste ALL of your code used to create your assignment including any RMarkdown code, data preprocessing code and text.
- Save the completed template file as a PDF or .doc/docx file. Name the file using your name and student number.
- Upload this assignment document by clicking "**Start Assignment**" above.

Note: You do not need to upload your data. However, you need to have it uploaded to Posit Cloud and your project ready to share with the marker if required.

Questions about Assignment 3? See [Assignment 3 - Discussion](#).

Failure to meet the submission requirements will incur a late penalty and delay your feedback.

Feedback will be provided using Canvas' Speed Grader™ and may include written inline feedback, general comments and/or audio feedback.

▼ 6. Assessment Criteria

A detailed marking rubric has been provided below.

▼ 7. Academic Integrity

Academic integrity is taken very seriously in this course. Put simply, when you submit assessment, it must be your own work (and you have not submitted that work for any other assessment/course). Therefore, this **prohibits the use of any generative "AI" tool** (e.g. ChatGPT) for completing assessment in this course. You will complete an Assessment declaration checklist in the assignment template to acknowledge your commitment to academic integrity. Assignments will be submitted through Turnitin, so if you've copied material, code and text from the web without appropriate attribution or from a fellow classmate or previous student, it will be flagged, and you will be investigated for academic misconduct. Any use or suspected use of AI will also be investigated and treated as potential plagiarism. Formal warnings and academic penalties apply to those found to have breached academic integrity. You should ensure you understand your responsibilities by reading the RMIT University website on [academic integrity](#). Ignorance is no excuse.

[Study Support Hub](#): Academic skills advisors and librarians will help you with researching your topic, finding information, improving your academic writing skills, academic integrity, effective learning strategies, statistics, and more!

[Academic Integrity/plagiarism](#): You can achieve academic integrity by honestly submitting work that is your own. Presenting work that fails to acknowledge other people's work within yours can compromise academic integrity.

[Special consideration](#): If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration.

For more information, please ask questions in [Assignment 3 - Discussion](#).

Criteria	Ratings						Pts
Web-hosted	2 Pts Completed The submission was successfully hosted online using one of the approved services. The assignment was accessible the first time it was opened by a marker.		1 Pts Partially completed The submission was hosted online using one of the approved services, however, the assignment didn't load successfully the first time. Once notified, the student was quick to fix the issue by themselves.		0 Pts Not completed The submission was not correctly hosted or accessible online using one of the approved services. The student was slow to respond to requests to fix hosting and/or required support to fix the hosting issue.		2 pts
Assignment Document Upload	2 Pts Completed Document submitted to Canvas included the student name, student number, URL to the web-hosted assignment and all the code used for the assignment.		1 Pts Partially completed Document submitted to Canvas was missing information.		0 Pts Not completed Document submitted to Canvas was missing major information, e.g. code.		2 pts
Constraints The submission adhered to the constraints imposed on the different types of assignments permitted.	2 Pts Adhered to The submission adhered to the constraints detailed in the assignment instructions.		1 Pts Partially adhered to The submission mostly adhered to the constraints detailed in the assignment instructions. However, some constraints were relaxed.		0 Pts Not adhered to The submission ignores one or more major constraints detailed in the assignment instructions.		2 pts
Reference to data source (s)	2 Pts Completed Data used for the assignment was appropriately referenced in the web-hosted assignment using APA Style 7th Ed.		1 Pts Partially completed The source(s) of the data was NOT appropriately referenced in the web-hosted assignment using APA Style 7th Ed. For example, the source was referenced using only a URL.		0 Pts Not completed A reference to the source data was missing in the web-hosted assignment.		2 pts
Data quality This is a rating of the quality of	4 Pts Excellent	3 Pts Good	2 Pts Needed improvement	1 Pts Poor	0 Pts Not acceptable		4 pts

Criteria	Ratings					Pts
the data sourced, processed and visualised for the assignment. This criterion considers the appropriateness, timeliness, and reliability of the data selected. It also considers the opportunities to present other data/sources that would improve the impact of the visualisation.	The data used for the assignment was appropriate, timely and obtained from a reputable source(s).	The data used for the assignment was largely appropriate, timely and obtained from a reputable source(s). However, some minor improvements were suggested.	There were one or more issues with the appropriateness, timeliness and/or reliability of the data source(s). Improvement was needed.	The data sourced were poorly suited and/or of questionable reliability.	The data are not appropriate for the visualisation objective.	
Captivation and narrative This is a rating of the extent to which the final assignment presents a compelling and memorable visualisation(s). This criterion rewards compelling data visualisations with clear objectives and visuals accessible to a wide audience. Narrative visualisation techniques are highly regarded.	12 to >9.0 Pts Excellent Narrative visualisation techniques are used in a highly effective manner to guide the audience through an engaging and compelling visual story using data visualisation. The assignment leaves a lasting impression.	9 to >6.0 Pts Good The assignment had a clear and practical objective. Data visualisation was used effectively to meet the objective in an accessible manner and leaves a lasting impression. Some narrative visualisation techniques are employed. However, minor improvements were needed.	6 to >3.0 Pts Needs improvement The assignment had a clear objective, but the impact was limited. The use of data visualisation was not capitalised on and instead the audience had to rely on other elements. Narrative visualisation techniques were limited. Overall, improvement was needed.	3 to >0.0 Pts Poor The assignment lacked a strong, practical objective, and missed the opportunity to use data visualisation to tell a compelling story. Narrative visualisation techniques were ineffective or non-existent.	0 Pts Not acceptable The objective of the data visualisation was trivial or unclear. A narrative was missing.	12 pts

Criteria	Ratings					Pts
Visual methods This is a rating of the overall quality and demonstrated mastery of the data visualisation techniques used in the assignment. This criterion takes into account the appropriateness of the methods and techniques used to visualise the data, support the overall narrative and avoid deception.	12 to >9.0 Pts Excellent The data visualisation techniques used in the assignment were appropriate in relation to the data and objective. The methods were accessible, avoided deception and demonstrated a high level of mastery of open-source data visualisation packages.	9 to >6.0 Pts Good The data visualisation techniques used in the assignment were mostly appropriate in relation to the data and objective. The methods were largely accessible, avoided deception and demonstrated a solid understanding of open-source data visualisation packages. However, some minor improvements were suggested.	6 to >3.0 Pts Needs improvement The data visualisation methods were appropriate, but other methods were better suited to the data and/or objective.	3 to >0.0 Pts Poor The data visualisation techniques used in the assignment were not appropriate and had issues with accessibility, deception and/or suitability to the data and objective.	0 Pts No appropriate The data visualisation techniques were inappropriate, deceptive and not suited to the data.	12 pts
Publication quality This is a rating of the overall publication quality of the web-hosted assignment. This criterion considers the appropriate use of colour, headings, text (spelling and grammar), fonts, lines and arrangement (see Chapter 1 –	4 Pts Excellent The assignment demonstrates superb attention to detail. Colour is used effectively and with restraint. Headings, text and fonts are clear, concise, and do not detract from the data.	3 Pts Good The assignment demonstrates good attention to detail. However, there were a few minor issues with colour, headings, text, lines and/or arrangement etc that need	2 Pts Needs improvement The assignment is not publication-ready. While some elements have been prepared for publication, multiple issues with appearance such as text, colour,	1 Pts Poor The assignment is not publication-ready and appears to resemble default output taken from data visualisation packages. It appears very little effort has been dedicated to	0 Pts No marks The assignment demonstrates no regard for the publication quality of the assignment and data visualisations.	4 pts

Criteria	Ratings					Pts
Publication Ready Data Visualisations).	Lines and overall arrangement of data visualisations support the overall narrative and help to guide the audiences' attention effectively. The overall assignment looks professional and highly polished.	further attention.	headings and arrangement remain. There is much room for improvement.	preparing the data visualisation and overall assignment for publication.		
Total points: 40						